

DBS - Lab 06: A ‘gentle’ Introduction to SQL Indexing.

|                                            |                                                                                                                                                            |        | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---|---|---|---|---|---|---|---|---|----|----|----|
| Criteria                                   | Description                                                                                                                                                | Points |   |   |   |   |   |   |   |   |   |    |    |    |
| Database Schema & Data Generation          | Successfully creates account, thread, post, and words tables. Populates 100k posts and 1k threads using the random generation scripts provided.            | 10     |   |   |   |   |   |   |   |   |   |    |    |    |
| Performance Analysis (Q2-Q5)               | Correctly executes EXPLAIN ANALYZE for all four queries (Q2-Q5). Provides clear "Before" and "After" execution time statistics.                            | 15     |   |   |   |   |   |   |   |   |   |    |    |    |
| Visualization 1: Before vs. After Indexing | Produces a clear bar chart using R, Python, Octave, or Gnuplot. Includes two bars per query on the X-axis and execution time on the Y-axis.                | 15     |   |   |   |   |   |   |   |   |   |    |    |    |
| Index Type Comparison (B-tree, Hash, BRIN) | Selects one query and measures performance across B-tree, Hash, and BRIN index types. Correctly implements the specific syntax for each (e.g., USING HASH) | 15     |   |   |   |   |   |   |   |   |   |    |    |    |
| Visualization 2: Index Type Performance    | Generates a secondary plot where the X-axis represents index types and the Y-axis shows execution time. Follows all formatting constraints.                | 15     |   |   |   |   |   |   |   |   |   |    |    |    |
| Report structure and clarity               | Submit a well-structured and clear report in PDF format, ensuring readability and proper formatting.                                                       | 10     |   |   |   |   |   |   |   |   |   |    |    |    |
| Lab Defense                                | Oral presentation or defense of the lab work, demonstrating understanding of concepts and decisions.                                                       | 20     |   |   |   |   |   |   |   |   |   |    |    |    |
| Total:                                     |                                                                                                                                                            | 100    |   |   |   |   |   |   |   |   |   |    |    |    |